Lecture 6. Jointly Distributed Random Variables

Joint and marginal laws, independence, convolutions, conditional laws, order statistics, Jacobians

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6.1 Joint distribution functions

The **joint cdf** of X, Y is $F(a,b) = P(X \le a, Y \le b)$; the marginals are $F_X(a) = F(a,\infty)$ and $F_Y(b) = F(\infty, b)$, and

$$P(a_1 < X \le a_2,\ b_1 < Y \le b_2) = F(a_2,b_2) - F(a_1,b_2) - F(a_2,b_1) + F(a_1,b_1).$$

For discrete variables the joint pmf is $p(x,y) = P(X = x, Y = y)$, and $p_X(x) = \sum_y p(x,y)$. X and Y are **jointly continuous** with joint density f if $P((X,Y) \in C) = \iint_C f(x,y)\,dx\,dy$; then $f = \partial^2 F/\partial a\,\partial b$, and $f_X(x) = \int f(x,y)\,dy$. All of this extends to n variables.

Example 6.1

Let $f(x,y) = 2e^{-x}e^{-2y}$ for $x, y > 0$. Then $P(X > 1, Y < 1) = \int_0^1 \int_1^\infty 2e^{-x}e^{-2y}\,dx\,dy = e^{-1}(1 - e^{-2}) \approx 0.318$, and

$$P(X < Y) = \int_0^\infty \int_0^y 2e^{-x}e^{-2y}\,dx\,dy = \int_0^\infty 2e^{-2y}(1 - e^{-y})\,dy = 1 - \tfrac{2}{3} = \tfrac{1}{3}.$$

Example 6.2 (A random point in a disc)

(X,Y) is uniform on the disc of radius R: $f = 1/(\pi R^2)$ there. The marginal is $f_X(x) = \frac{2\sqrt{R^2 - x^2}}{\pi R^2}$ for $|x| \le R$, which is not uniform. The distance D to the centre has $P(D \le a) = \pi a^2/(\pi R^2) = a^2/R^2$, so $f_D(a) = 2a/R^2$ and $E[D] = 2R/3$.

Example 6.3 (Multinomial distribution)

In n independent trials, each resulting in outcome i with probability p_i ($\sum_1^r p_i = 1$), the counts $X_1, \dots, X_r$ satisfy

$$P(X_1 = n_1, \dots, X_r = n_r) = \frac{n!}{n_1! \cdots n_r!} p_1^{n_1} \cdots p_r^{n_r}, \qquad \sum n_i = n,$$

since each sequence with these counts has probability $\prod p_i^{n_i}$ and there are $\binom{n}{n_1,\dots,n_r}$ such sequences (Lecture 1). Each X_i is marginally $\mathrm{Bin}(n, p_i)$.

6.2 Independent random variables

Motivation. Independence means the joint distribution factors into marginals. It is what makes joint probabilities computable and underlies sums, products and the limit theorems.

X and Y are **independent** if $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for all A, B; equivalently $F(a,b) = F_X(a)F_Y(b)$, or $p(x,y) = p_X(x)p_Y(y)$, or $f(x,y) = f_X(x)f_Y(y)$.

Proposition 6.1 (Factorization)

Jointly continuous (or discrete) X, Y are independent iff the joint density (pmf) factors as $f(x,y) = h(x)g(y)$ for all x, y.

Proof. If $f = hg$, put $C_1 = \int h$, $C_2 = \int g$; then $C_1C_2 = \iint f = 1$, $f_X = hC_2$ and $f_Y = gC_1$, so $f_X f_Y = hg\,C_1C_2 = f$. ∎

The factorization must hold on the whole plane: $f = 24xy$ on $\{x, y > 0, x + y < 1\}$ looks like a product but is not, because the support is not a rectangle.

Example 6.4 (Thinning a Poisson count)

The number N of customers on a day is Poisson(λ), and each is independently a man with probability p. Let X and Y be the numbers of men and women. Then

$$P(X = i, Y = j) = e^{-\lambda}\frac{\lambda^{i+j}}{(i+j)!}\binom{i+j}{i}p^i q^j = e^{-\lambda p}\frac{(\lambda p)^i}{i!}\cdot e^{-\lambda q}\frac{(\lambda q)^j}{j!}:$$

X and Y are *independent* Poisson variables with means λp and λq. (With a fixed number n of customers, they would be strongly dependent.)

Example 6.5 (Meeting)

Two people arrive independently and uniformly between 12:15 and 12:45 and each waits 10 minutes. Measuring in minutes past 12:15, they meet iff $|X - Y| \le 10$, so $P = 1 - (20/30)^2 = 5/9$ (the two corner triangles of the square have total area 20^2).

Example 6.6 (Buffon's needle)

A needle of length L is dropped on a floor ruled with parallel lines $D \ge L$ apart. Let $X \sim U(0, D/2)$ be the distance from the needle's centre to the nearest line and $\theta \sim U(0, \pi/2)$ the acute angle with the lines, independent. The needle crosses a line iff $X < \frac{L}{2}\cos\theta$, so

$$P(\text{cross}) = \frac{4}{\pi D}\int_0^{\pi/2}\int_0^{(L/2)\cos\theta} dx\, d\theta = \frac{2L}{\pi D}.$$

Dropping many needles thus estimates π.

Example 6.7 (Characterizing the normal)

Darts land at (X, Y) with X, Y independent with a common differentiable density f, and the joint density depends only on the distance from the target: $f(x)f(y) = g(x^2 + y^2)$. Differentiating in x gives $f'(x)f(y) = 2xg'(x^2 + y^2)$, so $\frac{f'(x)}{2xf(x)} = \frac{g'(x^2+y^2)}{g(x^2+y^2)}$ for all x, y. The left side does not depend on y, and the right side then shows it does not depend on x; hence $f'(x)/f(x) = cx$, $f(x) = ke^{cx^2/2}$ with $c < 0$: X is normal with mean 0.

6.3 Sums of independent random variables

Motivation. The distribution of a sum of independent variables is the convolution of their distributions. Several families are closed under sums (normal, Poisson, gamma), which explains their importance.

If X, Y are independent with densities f_X, f_Y, then

$$F_{X+Y}(a) = \int F_X(a - y)f_Y(y)\, dy, \qquad f_{X+Y}(a) = \int f_X(a - y)f_Y(y)\, dy,$$

the **convolution** of f_X and f_Y.

Example 6.8 (Sums of uniforms)

For $X, Y \sim U(0,1)$ independent, $f_{X+Y}(a) = a$ on $[0,1]$ and $2-a$ on $[1,2]$ (triangular). More generally, if S_n is a sum of n independent uniforms, induction on n gives $P(S_n \le x) = x^n/n!$ for $0 \le x \le 1$. Let $N = \min\{n : S_n > 1\}$. Then $P(N > n) = P(S_n \le 1) = 1/n!$, so

$$E[N] = \sum_{n \ge 0} P(N > n) = \sum_{n \ge 0} \frac{1}{n!} = e.$$

Proposition 6.2

For independent variables:
(a) $\text{Gamma}(s,\lambda) + \text{Gamma}(t,\lambda) \sim \text{Gamma}(s+t,\lambda)$; in particular a sum of n independent $\text{Exp}(\lambda)$ is $\text{Gamma}(n,\lambda)$ and $Z_1^2 + \cdots + Z_n^2 \sim \chi^2_n$.
(b) $\sum N(\mu_i, \sigma_i^2) \sim N(\sum \mu_i, \sum \sigma_i^2)$.
(c) $\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) \sim \text{Poisson}(\lambda_1 + \lambda_2)$.
(d) $\text{Bin}(n,p) + \text{Bin}(m,p) \sim \text{Bin}(n+m,p)$.

Proof. (a) The convolution is $\frac{\lambda e^{-\lambda a}}{\Gamma(s)\Gamma(t)} \lambda^{s+t-1} \int_0^a y^{t-1}(a-y)^{s-1}\,dy$; substituting $y = ax$ shows it is $Ca^{s+t-1}e^{-\lambda a}$, a gamma density, and C is then forced.

(b) It suffices to treat $N(0,\sigma^2)+N(0,1)$, a direct completion of the square; Lecture 7 gives a one-line proof with moment generating functions.

(c) $P(X+Y=n) = \sum_k e^{-\lambda_1}\frac{\lambda_1^k}{k!} e^{-\lambda_2}\frac{\lambda_2^{n-k}}{(n-k)!} = e^{-(\lambda_1+\lambda_2)}\frac{(\lambda_1+\lambda_2)^n}{n!}$ by the binomial theorem.

(d) Count successes in $n+m$ trials, or use Vandermonde. ∎

Example 6.9

A team plays 26 games against class A teams, winning each with probability 0.4, and 18 against class B, winning each with probability 0.7, all independently. Its number of wins has mean $26(0.4) + 18(0.7) = 23$ and variance $26(0.24) + 18(0.21) = 10.02$; it is not binomial, but the normal approximation applies: $P(\text{wins} \ge 25) \approx 1 - \Phi(1.5/\sqrt{10.02}) \approx 0.32$.

6.4 Conditional distributions

For discrete variables, $p_{X|Y}(x \mid y) = p(x,y)/p_Y(y)$ when $p_Y(y) > 0$. For jointly continuous ones, the conditional density is

$$f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)}, \qquad P(X \in A \mid Y = y) = \int_A f_{X|Y}(x \mid y)\,dx,$$

which is motivated by $P(x < X < x+dx \mid y < Y < y+dy) \approx f(x,y)\,dx\,dy/(f_Y(y)\,dy)$. Independence means the conditional law does not depend on y.

Example 6.10

(a) If X, Y are independent Poisson with means λ_1, λ_2, then given $X + Y = n$, $X \sim \text{Bin}(n, \lambda_1/(\lambda_1+\lambda_2))$. Likewise, given $X + Y = n$ for independent $\text{Bin}(n_1, p)$ and $\text{Bin}(n_2, p)$, X is hypergeometric, free of p.
(b) In a multinomial, given $X_r = n_r$, the remaining counts are multinomial with $n - n_r$ trials and probabilities $p_i/(1-p_r)$.

(c) If $f(x,y) = e^{-x/y}e^{-y}/y$ for $x, y > 0$, then $f_{X|Y}(x \mid y) = \frac{1}{y}e^{-x/y}$: given $Y = y$, X is exponential with mean y, and $P(X > 1 \mid Y = y) = e^{-1/y}$.

Example 6.11 (The bivariate normal)

(X, Y) is bivariate normal with parameters $\mu_x, \mu_y, \sigma_x, \sigma_y, \rho$ ($|\rho| < 1$) if

$$f(x,y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \exp\Big\{-\frac{1}{2(1-\rho^2)}\Big[u^2 - 2\rho uv + v^2\Big]\Big\}, \quad u = \frac{x-\mu_x}{\sigma_x},\ v = \frac{y-\mu_y}{\sigma_y}.$$

Completing the square in u shows that, given $Y = y$, X is normal with

$$E[X \mid Y = y] = \mu_x + \rho\frac{\sigma_x}{\sigma_y}(y - \mu_y), \qquad \mathrm{Var}(X \mid Y = y) = \sigma_x^2(1-\rho^2),$$

and integrating out x shows $Y \sim N(\mu_y, \sigma_y^2)$. X and Y are independent iff $\rho = 0$ (Lecture 7 shows ρ is the correlation).

Example 6.12 (Mixed variables: beta–binomial)

Let $P \sim U(0,1)$ and, given $P = p$, let $N \sim \mathrm{Bin}(n,p)$. Then $P(N = k) = \int_0^1 \binom{n}{k} p^k(1-p)^{n-k}\,dp = \binom{n}{k}B(k+1, n-k+1) = \frac{1}{n+1}$: the number of successes is uniform on $\{0, \dots, n\}$. Given $N = k$, the density of P is proportional to $p^k(1-p)^{n-k}$: $P \mid N = k \sim \mathrm{Beta}(k+1, n-k+1)$. Compare Laplace's rule of succession (Example 3.12).

6.5 Order statistics

Let $X_1, \dots, X_n$ be i.i.d. with density f and cdf F, and $X_{(1)} \le \cdots \le X_{(n)}$ their ordered values. Each of the $n!$ assignments of the X_i to the ordered values is equally likely, so

$$f_{X_{(1)},\dots,X_{(n)}}(x_1, \dots, x_n) = n!\, f(x_1)\cdots f(x_n), \qquad x_1 < \cdots < x_n.$$

For one order statistic: $X_{(j)}$ is near x when $j-1$ observations are below x, one is near x, and $n-j$ are above, so

$$f_{X_{(j)}}(x) = \frac{n!}{(j-1)!\,(n-j)!} F(x)^{j-1}(1-F(x))^{n-j} f(x).$$

Also $P(X_{(j)} \le x) = \sum_{k=j}^{n} \binom{n}{k} F(x)^k(1-F(x))^{n-k}$ (at least j observations below x). For uniforms, $X_{(j)} \sim \mathrm{Beta}(j, n-j+1)$, so $E[X_{(j)}] = j/(n+1)$.

Example 6.13

(a) Three people are placed independently and uniformly on a road of length 1. The probability that no two are within distance $d \le \frac{1}{2}$ is

$$3! \int_{x_1 < x_2 - d,\ x_2 < x_3 - d} dx_1\, dx_2\, dx_3 = (1-2d)^3,$$

by substituting $y_1 = x_1$, $y_2 = x_2 - d$, $y_3 = x_3 - 2d$, which maps the region onto the ordered simplex in $[0, 1-2d]^3$.

(b) The sample median of $2n+1$ uniforms has density $\frac{(2n+1)!}{n!\,n!}x^n(1-x)^n$, and the range $R = X_{(n)} - X_{(1)}$ has $P(R \le a) = n\int f(x)\{F(x+a) - F(x)\}^{n-1}\,dx$; for uniforms, $f_R(a) = n(n-1)a^{n-2}(1-a)$.

6.6 Functions of random variables: the Jacobian method

Motivation. Changing variables in a joint density works like change of variables in a double integral: the Jacobian determinant accounts for the stretching of area. This yields, for instance, the polar method for normals.

Theorem 6.3

Let (X_1, X_2) have density f, and let $y_1 = g_1(x_1, x_2)$, $y_2 = g_2(x_1, x_2)$ be a one-to-one differentiable map with inverse $x_i = h_i(y_1, y_2)$ and nonzero Jacobian $J(x_1, x_2) = \det(\partial g_i / \partial x_j)$. Then $Y_i = g_i(X_1, X_2)$ have joint density

$$f_{Y_1,Y_2}(y_1, y_2) = f\big(h_1(y), h_2(y)\big)\, |J(h_1(y), h_2(y))|^{-1}.$$

This is the change of variables formula for double integrals; it holds in n dimensions as well.

Example 6.14 (Polar coordinates of normals)

Let X, Y be independent $N(0,1)$, and $R = \sqrt{X^2 + Y^2}$, $\Theta = \arctan(Y/X) \in [0, 2\pi)$. The Jacobian of $(x, y) \mapsto (r, \theta)$ is $1/r$, so

$$f_{R,\Theta}(r, \theta) = \frac{1}{2\pi} e^{-r^2/2}\, r = \frac{1}{2\pi} \cdot r e^{-r^2/2}:$$

R and Θ are independent, Θ is uniform, and $R^2 \sim \mathrm{Exp}(1/2) = \chi^2_2$. Reversing this gives the Box–Muller method: $X = \sqrt{-2\log U_1}\cos(2\pi U_2)$ and $Y = \sqrt{-2\log U_1}\sin(2\pi U_2)$ are independent standard normals.

Example 6.15 (Gamma and beta)

Let $X \sim \mathrm{Gamma}(\alpha, \lambda)$ and $Y \sim \mathrm{Gamma}(\beta, \lambda)$ be independent, $U = X + Y$, $V = X/(X+Y)$. Then $x = uv$, $y = u(1-v)$, $|J|^{-1} = u$, and

$$f_{U,V}(u, v) = \frac{\lambda e^{-\lambda u}(\lambda u)^{\alpha+\beta-1}}{\Gamma(\alpha+\beta)} \cdot \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} v^{\alpha-1}(1-v)^{\beta-1}.$$

So $U \sim \mathrm{Gamma}(\alpha+\beta, \lambda)$, $V \sim \mathrm{Beta}(\alpha, \beta)$, and they are independent; as a by-product $B(\alpha, \beta) = \Gamma(\alpha)\Gamma(\beta)/\Gamma(\alpha+\beta)$. For instance, if two jobs with exponential durations are done in sequence, the fraction of time spent on the first is uniform and independent of the total.

Example 6.16 (Spacings of exponentials)

For i.i.d. $\mathrm{Exp}(\lambda)$ variables $X_1, \dots, X_n$, the spacings $Y_1 = X_{(1)}$, $Y_j = X_{(j)} - X_{(j-1)}$ are independent with $Y_j \sim \mathrm{Exp}((n-j+1)\lambda)$. Indeed the map from ordered values to spacings has Jacobian 1, and $\sum_j x_{(j)} = \sum_j (n-j+1) y_j$, so the joint density $n!\lambda^n e^{-\lambda \sum x_{(j)}}$ factors. Intuitively, after the first failure the $n-1$ survivors start afresh by memorylessness. Hence $E[X_{(n)}] = \frac{1}{\lambda}\sum_{k=1}^{n} \frac{1}{k}$.

6.7 Exchangeable random variables

$X_1, \dots, X_n$ are **exchangeable** if $(X_{i_1}, \dots, X_{i_n})$ has the same joint law for every permutation. I.i.d. variables are exchangeable, but so are dependent ones such as the indicators X_i that the i-th

ball drawn without replacement is white: any particular sequence with k whites has probability $\frac{m(m-1)\cdots(m-k+1)\,(N-m)\cdots(N-m-n+k+1)}{N(N-1)\cdots(N-n+1)}$, which depends only on k. So every draw is white with probability m/N, as in Example 2.1(b). Pólya's urn (draw a ball, return it with another of the same colour) also gives exchangeable draws.

6.8 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 6 (book numbering). They practise the three basic operations on joint laws: integrating out to get the law of a function, factorizing a survival function, and conditioning by Bayes's rule in its density form.

Exercise 6.5 (Ratio and product of independent variables)

Let X and Y be independent positive continuous random variables with densities f_X and f_Y.
(a) Show that $Z = X/Y$ has density $f_Z(z) = \int_0^\infty y\, f_X(zy) f_Y(y)\, dy$, $z > 0$.
(b) Show that $W = XY$ has density $f_W(w) = \int_0^\infty \frac{1}{y} f_X(w/y) f_Y(y)\, dy$, $w > 0$.
(c) Evaluate f_Z when $X \sim \mathrm{Exp}(\lambda)$ and $Y \sim \mathrm{Exp}(\mu)$.

Solution.

(a)

Step 1. By independence the joint density is $f_X(x) f_Y(y)$ (Proposition 6.1). For $z > 0$,

$$F_Z(z) = P(X \le zY) = \int_0^\infty \Big(\int_0^{zy} f_X(x)\, dx \Big) f_Y(y)\, dy.$$

Step 2. In the inner integral substitute $x = uy$ ($y > 0$ fixed), so $dx = y\, du$ and u runs over $(0, z)$. Since the integrand is nonnegative, the order of integration may be exchanged (Tonelli):

$$F_Z(z) = \int_0^z \Big(\int_0^\infty y\, f_X(uy) f_Y(y)\, dy \Big)\, du.$$

Hence F_Z is the integral of the bracket, which is therefore a density of Z.

(b) In the same way, $F_W(w) = \int_0^\infty \int_0^{w/y} f_X(x)\, dx\, f_Y(y)\, dy$; substitute $x = u/y$, $dx = du/y$, and exchange the integrals to get $F_W(w) = \int_0^w \int_0^\infty \frac{1}{y} f_X(u/y) f_Y(y)\, dy\, du$.

(c) By (a), for $z > 0$,

$$f_Z(z) = \int_0^\infty y\, \lambda e^{-\lambda z y} \mu e^{-\mu y}\, dy = \frac{\lambda\mu}{(\lambda z + \mu)^2},$$

using $\int_0^\infty y e^{-ay}\, dy = 1/a^2$ with $a = \lambda z + \mu$. ■

Exercise 6.8 (Hazard rate of a minimum)

Let X and Y be independent positive continuous random variables with hazard rates λ_X and λ_Y, and let $W = \min(X, Y)$.
(a) Express the distribution function of W through F_X and F_Y.
(b) Show that $\lambda_W(t) = \lambda_X(t) + \lambda_Y(t)$.

Solution.

(a) $W > t$ iff $X > t$ and $Y > t$, so by independence $\bar F_W(t) = \bar F_X(t) \bar F_Y(t)$, i.e. $F_W(t) = 1 - \{1 - F_X(t)\}\{1 - F_Y(t)\}$.

(b)

Step 1. At every t where f_X and f_Y are continuous, $\bar{F}'_X = -f_X$ and $\bar{F}'_Y = -f_Y$, so by the product rule

$$f_W(t) = -\bar{F}'_W(t) = f_X(t)\bar{F}_Y(t) + \bar{F}_X(t)f_Y(t).$$

Step 2. Dividing by $\bar{F}_W(t) = \bar{F}_X(t)\bar{F}_Y(t) > 0$ gives $\lambda_W(t) = f_X(t)/\bar{F}_X(t) + f_Y(t)/\bar{F}_Y(t) = \lambda_X(t) + \lambda_Y(t)$.

In particular the minimum of independent $\mathrm{Exp}(\lambda)$ and $\mathrm{Exp}(\mu)$ has constant hazard $\lambda + \mu$, so it is $\mathrm{Exp}(\lambda + \mu)$: competing risks add. ■

Exercise 6.14 (Two geometric waiting times given their sum)

Let X and Y be independent geometric random variables with the same parameter p. Find $P(X = i \mid X + Y = n)$.

Solution.

Step 1. (Guess.) Flip a p-coin repeatedly; X is the time of the first head and $X + Y$ that of the second. Given that the second head is on flip n, the first head should be equally likely to be on any of flips $1, \dots, n-1$.

Step 2. (Verification.) With $q = 1 - p$, for $n \geq 2$ and $1 \leq i \leq n-1$, independence gives

$$P(X = i,\, Y = n - i) = q^{i-1}p \cdot q^{n-i-1}p = p^2 q^{n-2},$$

which does not depend on i. Summing over the $n-1$ values of i, $P(X + Y = n) = (n-1)p^2q^{n-2}$, so

$$P(X = i \mid X + Y = n) = \frac{1}{n-1}, \qquad i = 1, \dots, n-1.$$

■

Exercise 6.21 (Poisson counts with a gamma rate)

The moisture W on a day is $\mathrm{Gamma}(t, \lambda)$, and given $W = w$ the number N of accidents that day is $\mathrm{Poisson}(w)$. Show that given $N = n$, W is $\mathrm{Gamma}(t + n, \lambda + 1)$.

Solution.

Step 1. For the mixed pair (W, N) the conditional density of W given $N = n$ is, by Bayes's formula in density form (as in Example 6.12),

$$f_{W|N}(w \mid n) = \frac{P(N = n \mid W = w) f_W(w)}{P(N = n)}, \qquad P(N = n) = \int_0^\infty P(N = n \mid W = w) f_W(w)\, dw.$$

Step 2. For $w > 0$,

$$P(N = n \mid W = w) f_W(w) = e^{-w}\frac{w^n}{n!} \cdot \frac{\lambda e^{-\lambda w}(\lambda w)^{t-1}}{\Gamma(t)} = C_n\, e^{-(\lambda+1)w} w^{t+n-1},$$

where $C_n = \lambda^t / \{n!\,\Gamma(t)\}$ does not depend on w.

Step 3. So $f_{W|N}(w \mid n)$ is proportional to $e^{-(\lambda+1)w}w^{t+n-1}$, which is the kernel of the $\mathrm{Gamma}(t + n, \lambda + 1)$ density. Two densities that are proportional are equal, since both integrate to 1. Hence $W \mid N = n \sim \mathrm{Gamma}(t + n, \lambda + 1)$; in particular $E[W \mid N = n] = (t + n)/(\lambda + 1)$. ■